SENTINEL

Designing a Structured Threat Intelligence Platform for Non-Consensual Explicit Imagery

Team Sentinel · IS 492 Capstone · Spring 2026
Sage Kim · Sammy Haskel · Qiming Li · Yu-Chen (Will) Su
GitHub: https://github.com/IS492-SP26/team-project-deepfakes
Demo: https://willsu42-extended-team-project-deep.vercel.app

Recap & Final Goals

Problem Statement

No existing tool offers structured, technical documentation of how NCEI attacks are built — creating a critical gap for researchers, legal advocates, and platform integrity teams who need actionable, machine-readable intelligence to defend against and litigate deepfake harms.

Target Users

Safety / AI researchers needing technical incident archives
Platform integrity & policy teams
Technology law practitioners
Academic researchers studying deepfake harms

Most Rewarding Finding

USE1 & USE2 were the top-rated items (5.6 and 5.8 / 7) — users confirmed Sentinel provides value that no existing tool offers. Taxonomy usefulness was immediately recognized by researcher participants without prompting.

From CP3 ⟶ CP4: Moved from prototype validation to full formative usability study (N=7) with SUS, UMUX-Lite, trust/satisfaction scales, and qualitative coding.

Evaluation Design Overview

Two-Part Evaluation Design

Part 1 — Prompting Experiments

3 AI platforms tested (ChatGPT, Perplexity, Microsoft Copilot)
Tasks replicating Sentinel's core workflows
Purpose: Validate competitive gap & design rationale
Latency comparison: 10 min manual → <2 sec automated

Part 2 — Formative Usability Study

N=7 participants (3 DS/AI students, 3 IS/CS students, 1 law student)
Moderated think-aloud, 7 tasks (T1–T7)
Mean session: ~55 minutes
No prior exposure to Sentinel prototype

Metrics Collected

Task Success
0 / 0.5 / 1.0 per task

Time-on-Task
Moderator-timed (seconds)

Error Rate
Wrong clicks & dead ends

SUS Score
System Usability Scale (0–100)

UMUX-Lite
Usability metric (0–100)

Trust/SAT
TR1–TR3, USE1–USE2, SAT (7-pt)

Qualitative: affinity mapping of think-aloud notes + semi-structured debrief interview transcripts

Study Materials & Protocol

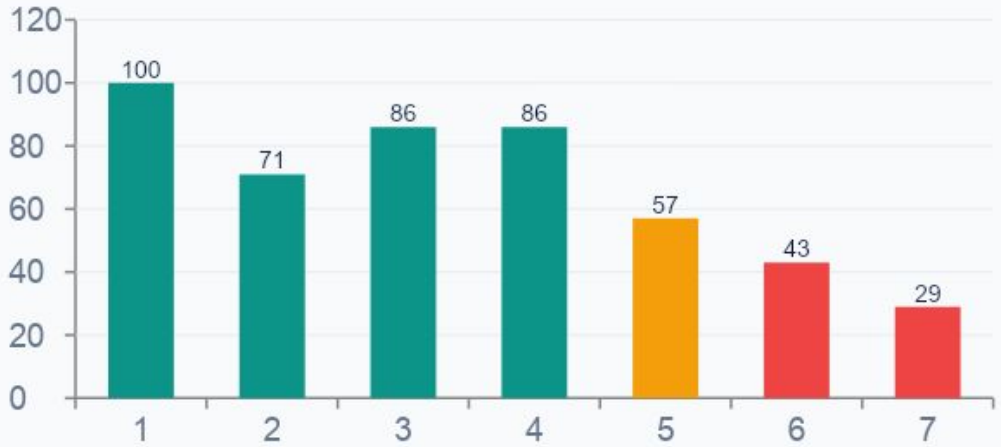
1. **Briefing & Consent** — 5 min
2. **Cold-Start Orientation** — 3 min
3. **Think-Aloud Tasks T1–T7** — 25–30 min
4. **Survey (SUS/UMUX/Trust)** — 10 min
5. **Debrief Interview** — 10 min

Task Set Overview (T1–T7)

Task	Description	Limit
T1	First-impression assessment	3 min
T2	Keyword search + category filter	4 min
T3	Structured metadata retrieval	4 min
T4	Incident Feed navigation	3 min
T5	Legal precedent lookup (TAKE IT DOWN Act)	4 min
T6	Frequency analysis across incidents	5 min
T7	Export / share feature discovery	3 min

Quantitative Results

Task Success Rates



← *80% benchmark target*

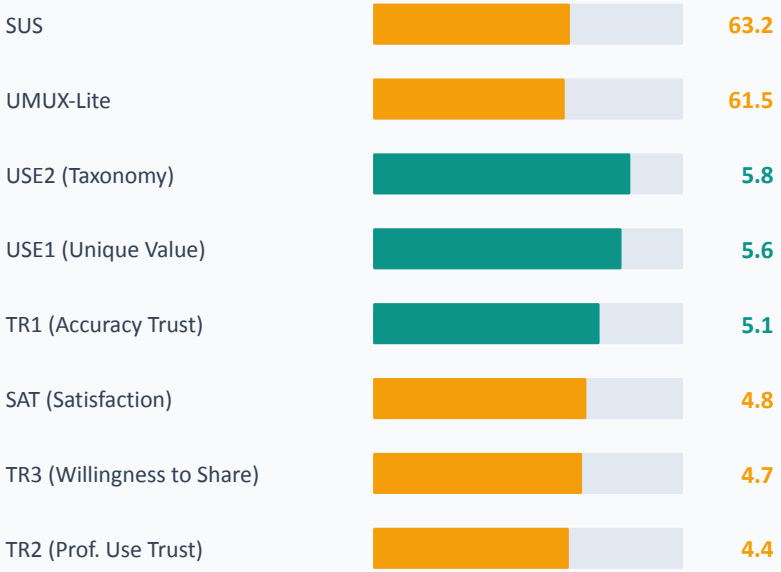
67%

Overall Task Success

153s

Median Time-on-Task

Survey Scores



Qualitative Insights

5 themes emerged from think-aloud notes & debrief interviews (N=7)

✔ Positive Themes

Taxonomy Value Recognized (High Freq.)

DS/AI students liked domain-specific vocab (LoRA fine-tuning, bypass method, prompt injection vector) as aligning well with how they think about threat modeling.

Strong First Impression (High Freq.)

T1 scored 100%: all 7 users correctly identified Sentinel's purpose within 90 seconds. Clinical View (raw article + structured metadata) was praised by those who discovered it.

✖ Frustrations & Failures

Navigation & Discoverability (High Sev.)

6/7 users failed to find category filter during T2. Filter is embedded in secondary panel, not as persistent sidebar. Legal Precedents section also hard to locate.

Missing Export Functionality (High Sev.)

T7 scored only 29%. 6/7 users expected CSV/JSON/Copy export; none found it. Three users brought up potential use cases: legal briefs, academic papers, threat databases.

> *"I know there should be a way to filter this, but I don't see it anywhere obvious."*
>
> — Participant, Task 2 (Category Filter)

Additional theme: Emotional/subject-matter response noted by 3 participants; no dropouts, but content warnings within records warranted.

Interpretation & Discussion

Core Finding: Users believe in Sentinel's unique value — but the interface isn't comfortable enough for professional use.

Usefulness ≠ Usability

USE scores (5.6–5.8/7) far outpace SUS (63.2) and UMUX-Lite (61.5). Expert-tool design challenge: specialists understand the concept but the interface isn't yet efficient for professional workflows.

Trust Gap Is Actionable

TR2 (professional-use willingness: 4.4/7) is the most consequential metric. Path to improvement: surface confidence scores, source citations, and human-verification status on every incident card.

Low-Scoring Tasks Are Feature Gaps

T5 (57%), T6 (43%), T7 (29%) partly reflect missing features (legal index UI, frequency aggregation, export), not only friction — directly informing the Version 2.0 roadmap.

Prompting Experiments Validated the Gap:

ChatGPT: Coarse-grained output, no JSON schema enforcement **Perplexity:** No central technical archive for model attribution
Copilot: Hard blocked — safety filters cannot distinguish forensic research from malicious intent **Manual latency:** >10 min/incident vs. Sentinel's <2 sec automated pipeline (99.7% reduction)

Limitations, Risks & Ethics

Evaluation Limitations

N=7 participants (academic environment)
Effect sizes not reliable at this scale
Directional results, not definitive
Live deployment may have load variability

System Limitations

Analyst-initiated only — non-celebrity cases under-covered
AI classification can produce inaccurate labels on borderline verdicts
Human review required for all results
Access controls not yet implemented

Misuse & Dual-Use Risk

Bypass technique documentation could be misused
Mitigation: researcher access restrictions (planned)
All queries logged; no step-by-step attack instructions stored

Data Privacy & Ethics

Victim metadata: category only (public figure / private individual / minor) — no identifying info
AI forensic context may require institutional governance
Forensic documentation rationale mirrors CVE databases in cybersecurity

Informed consent obtained; no explicit images shown; content warnings provided; study conducted per academic human subjects norms.

Conclusion & Future Work

Key Takeaways

1. The NCEI threat intelligence gap is an information architecture problem, not just a data problem — existing tools log that harm occurred; Sentinel logs how.

2. Users confirmed they could not find this type of structured information anywhere else (USE1: 5.6/7), and found the specific metadata fields — model used, bypass method, and takedown time — genuinely useful for their work (USE2: 5.8/7). This directly validates Sentinel's core purpose: that breaking down *how* NCEI attacks happen, not just *that* they happened, is something the field needs and currently lacks.

"Sentinel's long-term vision: the CVE database equivalent for NCEI — structured, authoritative, self-updating."

Version 2.0 Roadmap

- Persistent filter sidebar (fix T2/T5 failures)
- CSV / JSON / citation-ready export pipeline
- Confidence scores + provenance badges on every card
- In-context glossary for legal/policy users
- Access control & researcher credentialing
- Court record APIs + FOIA integration
- Longitudinal incident tracking (living records)

Acknowledgments & Contributions

Individual Responsibilities

Sage Kim	User study design, qualitative analysis, affinity mapping, IRB coordination
Sammy Haskel	Frontend development (React/Vite), UX design, usability task design
Qiming Li	Tech tools research , database schema, API Testing
Yu-Chen (Will) Su	Full-stack development, deployment (Vercel), AI integration (Claude/Groq), system architecture

AI Tools & Resources Disclosure

Tool	Usage / Disclosure
Claude (Anthropic)	Used to reformat and refactor code and written text for readability, professionalism, and interpretability throughout project development. Also used as a dual AI backend (Claude Sonnet 4) for MIT Causal Taxonomy classification.
Gemini (Google)	Used to reformat and refactor code and written text for readability, professionalism, and interpretability during report writing and documentation phases.
ChatGPT / GPT-4o	Used in prompting experiments (§4.2, Case 1) to validate the competitive gap in structured metadata extraction.
Perplexity.ai	Used in prompting experiments (§4.2, Case 2) to validate the absence of a central technical archive for NCEI incidents.
Groq / Llama 3.3-70b	Integrated as a second AI backend option within the Sentinel platform for taxonomy classification.